

Opening

Respected Judges and Industry Leaders,

Kya aapko pata hai ki jab aap ek standard ₹10 ka biscuit packet kharidte hain, toh uspe likhi **font size ya missing unit price** ki wajah se har saal India mein kitna revenue leak hota hai? ₹1,800 Crore+ per year!

Aaj India ke 3,000+ Legal Metrology Officers millions of retail shops aur godowns ko inspect karte hain. Lekin problem kya hai?

Manual Friction: Ek single package ki compliance check karne mein 20 se 25 minutes lagte hain—vernier calipers aur magnifying glass ke sath. Result? Ek officer din mein mushkil se 15–20 products hi inspect kar paata hai.

The Legal Void: July 1, 2024 ko Section 65B repeal hone ke baad, under Section 63 of Bharatiya Sakshya Adhiniyam (BSA) 2023, normal digital photos court mein reject ho jaati hain. Result? 35% dismissals in court!

We asked ourselves: Why are we using 19th-century manual tools for 21st-century enforcement?

Namaste, we are Team CYPHER, and we present NIRIKSHAK—Automated Legal Metrology Verification & Evidentiary Vault.

NIRIKSHAK converts a slow 25-minute manual check into a sub-30-second digital workflow, ensuring 100% court admissibility and zero legal loopholes!"-----

Tech

Now the biggest question judges ask us: 'AI hallucinations se false legal notices generate nahi honge?'

Our answer is NO. NIRIKSHAK is built on a Deterministic Statutory AST Architecture. AI ka role sirf text read karne tak restricted hai, whereas legal math rules 100% Gazette-codified hain. Zero hallucinations, exact legal certainty!

Let me walk you through our 5-stage Technical Pipeline:

Sub-millimeter Optical Metrology:

Hum Planar Homography use karte hain. Inspector bas apna standard PAN/Aadhaar card as a fiducial scale product ke sath rakhta hai. Our system un-tilts perspective up to 35° and

measures font $\times$ -height with $\pm 0.08\text{mm}$ sub-millimeter precision—verified on physical digital calipers!

Multilingual Edge Vision: Powered by DBNet++ for polygon detection and PP-OCrv4 running quantized INT8 ONNX models, reading English and Devanagari Hindi in just 180ms on standard CPUs.

Zero-Cost Air-Gapped Topology: Remote rural mandis mein internet nahi hota. NIRIKSHAK works in Dual Mode: Mode A for HQ Cloud Sync, and Mode B (100% Offline PWA with local SQLite). And the best part for investors? ₹0 Recurring Cloud GPU Bill!
Section 63 BSA 2023

Evidentiary Vault: Every capture is locked with a SHA-256 byte digest, monotonic UTC timestamp, and anchored into a Cryptographic Merkle DAG, generating a tamper-proof, court-admissible Form-1 PDF/A certificate."

Ab baat karte hain feasibility, economic viability, aur national impact ki:

94% Time Reduction: Per-package inspection time 8–10 minutes se ghat kar under 30 seconds reh jata hai.

13x Daily Throughput: Per-officer daily capacity 15 SKUs se badh kar 200+ SKUs ho jati hai.

100% Court *Admissibility*: Section 63 BSA 2023 ke tehat Merkle DAG Root ke saath 1-click Form-1 Court Notice PDF/A generate hota hai, jisse district courts mein zero case dismissals honge.

Feasibility & ROI: Zero hardware upgrades required—yeh inspectors ke existing smartphones par chalta hai. Commercial LLM APIs par zero dependence ki waja se recurring cloud expense ₹0 hai, jabki yeh deceptive packaging dwara hone wale ₹1,800+ Crore ke state revenue leakage ko plug karega.

Phased Rollout: Phase 1 mein 50 District HQs pilot se lekar Phase 2 mein 3,000 field officers aur Phase 3 mein DoCA eMaap portal aur NCH integration ka clear roadmap tayar hai.

Conclusion

To conclude: NIRIKSHAK legal metrology ko physical vernier caliper disputes se nikaal kar AI-powered precision aur tamper-proof digital sovereignty ki taraf le jaata hai.

Thank you